

BIOGENIC PRIMARY SULFIDE OF THE PERMIAN KUPFERSCHIEFER AND MARL SLATE

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ABSTRACT

The sulfide of the Kupferschiefer of Germany includes myriads of small spheres mostly up to 20 μ in diameter. These are now shown to contain and be enclosed in delicate microfossil remains associated with *Pyritosphaera barbaria* Love 1957, which has been recognized in the pyrite microspheres of many dark shale rocks and some recent sediments. The principal forms are regarded as the remains of micro-organisms that caused the authigenic precipitation of the primary sulfide in the sediment of the foul-bottomed early Zechstein sea of Europe. Similar observations have been made from the Marl Slate of north east England, a contiguous deposit of the same sea. This primary sulfide of the Kupferschiefer has been shown by studies of other workers on pseudomorphism to have been pyrite and not of mixed sulfides including copper as postulated by earlier syngeneticists. Demonstration here of its early formation allows consideration of early diagenetic replacement and the current theories of the origin of this secondary mineralization are discussed in this light.

INTRODUCTION

THIS paper arises from paleontological studies of samples of the Permian Kupferschiefer of Mansfeld, Germany, and the equivalent Marl Slate from north east England. It gives in particular an explanation of the origin of the disseminated sulfide spheres abundantly occurring in them and thereby provides a contribution to the understanding of the metalliferous deposits of these rocks. The origin of the spheres has already engaged the attention of many writers without satisfactory conclusion but the approach and methods described in the present work give a new means of resolving the problem.

The forms in which sulfides of the *Speise* or disseminated ore of the

Kupferschiefer occur, especially the microscopic grains, have been described by Schneiderhöhn (20, 21) for the Mansfeld deposits and by Neuhaus (15) for those of the Kupfermergel of Lower Silesia. That the Marl Slate of north east England, briefly described by Dunham (4, p. 262-3) in its setting as an ore-bearing deposit, and the Kupferschiefer and the Kupfermergel are equivalent is discussed by Deans (3, p. 344-347) who described similar mineralization there. Concerning the form of the sulfide, the composition of which will be discussed later, Neuhaus' observations adequately sum up those of Schneiderhöhn and cover three groups.

(a) Most important in the present study are the *Kieskügelchen* (15, p. 318, Fig. 5). These well known small spheres 3 to 20 μ in diameter, are composed of closely packed sulfide particles each $\frac{1}{2}$ to 3 μ in size. The abundance of these spheres in the Kupferschiefer is notable. Both Schneiderhöhn (20, p. 14) and Schouten (22, p. 376) record 10,000 or more per cu mm of rock.

(b) Another characteristic sulfide form is that of the *Kiesklümpchen*. At 20 to 40 μ the size is greater than that of the *Kieskügelchen* and the individual grains, too, are coarser, generally about 3 to 8 μ across. Neuhaus (15, p. 322) notes their rarity in the Kupferschiefer of Mansfeld and although the present author has found the explanation of their form in other rocks (12), they hardly enter into the present paper.

(c) Individual crystals of the size and form of the grains from which the *Kieskügelchen* are built up are also found. Both Schneiderhöhn (20, p. 15) and Neuhaus (15, p. 322) refer to this as fine sulfide "dust" scattered throughout the bituminous groundmass of the shale.

The early formation of sulfide spheres, at least of pyrite, in the history of the Kupferschiefer is acceptable to all authors. Schneiderhöhn (20, 21) introduced the concept of *vererzte Bakterien* or mineralized bacteria to explain their origin syngenetic with deposition of the shale. As *vererzte Bakterien* the spheres were held to be the relics of sulfur bacteria whose sulfur had become fixed by a mixture of metals, including copper, previously in solution in the sea water. The great abundance of the spheres and their regular shape were significant factors in association with the obviously foul bottom nature of the deposits where sulfur bacteria would be expected to proliferate. To Schouten (9, 23) this theory was unacceptable, one of the grounds being the lack of resemblance of the spheres to any known sulfur bacteria. By detailed studies of pseudomorphism (22) the pyrite spheres were shown to be a phase generally replaced by other minerals—copper, zinc and lead sulfides—but, even by inference from studies of the Mount Isa Shale of Australia, Schouten was unable to disprove, conclusively, the primary syngenetic origin of the spheres as postulated by Schneiderhöhn. If indeed they were primary then probably Schouten would have regarded them as the product of the crystallization of small inorganically accumulated blobs of sulfide gel, as did Barringer (Taylor, in 11, p. 438) and Rust (19) for similar forms from recognized hypogene ore deposits.

By the broad range of rocks over which Schouten's discussion ranged it is clear how widespread in ore deposits is the occurrence of sulfide spheres.

Deans (3, p. 349–350) has noted that pyrite spheres are no less common in various coals and shale and are to be regarded, in fact, as characteristic of sapropelic rocks. His comment on their passive role in the mineralization of the Kupferschiefer in no way reduces their importance, in their enormous abundance, as hosts to many ore minerals found as pseudomorphs of the spheres after pyrite or marcasite.

To this general problem it has been relevant to apply the methods employed by the author on other rocks, a brief account of which follows. Pyrite spheres, of a size and type comparable with those of the Kupferschiefer were observed by the author in shales of the Lower Carboniferous (Mississippian) succession in Scotland. It was found possible (11) to isolate these pyrite spheres: removal of the pyrite from them by nitric acid left microfossil remains of an organic material, principally as small spheres, which had been contained within the pyrite spheres. Subsequent work has confirmed that in other deposits, such as the Lower Jurassic Shales of England (12) the same phenomenon can be demonstrated while a clearer idea of the morphology of the micro-organisms was obtained. It was held that the consistent association of these organisms with sulfide is due to their original generation of hydrogen sulfide in the mud possibly in the manner of the sulfate-reducing bacteria. As a result, acting also as the focus for the inevitable precipitation of iron from solution by the sulfide, they became infilled with some form of iron sulfide, perhaps hydrated, which ultimately crystallized as pyrite. No doubt iron sulfide also formed elsewhere as minute crystals in the mud. The organisms were thought to represent part, at least, of an organic life cycle and perhaps to be saprophytic in nature. Later studies, still continuing, suggest a simpler interpretation of their morphology, more consistent with that of bacteria. If indeed the metabolism of the organisms is correctly interpreted, then it is possible that they are representatives of the group *Desulfovibrio desulfuricans* Beijerinck (Kluyver and Van Niel). The long geological history of the phenomenon was demonstrated by Love and Zimmerman (14) who showed its prevalence in the Lower Proterozoic Mount Isa black shale sediments where, in fact, the organic form of sulfide accounts for enormous quantities of primary pyrite spheres and indeed represents a great source of pyrite available for mobilization into massive deposits.

In the present study, the Kupferschiefer specimens used were kindly released from the collections of the University of Freiberg i. Breisgau (Department of Geology), the Imperial College of Science & Technology and Royal School of Mines, London (Department of Mining Geology) and the author's own Department of Geology in the University of Sheffield. Carefully located underground material was thus available from the horizons of the Schwarzerberge, the Dachberge, the Kammschale and the Grosse and Feine Lette of the Kupferschiefer (21, p. 143 & Fig. 50). Specimens of Marl Slate from north east England were kindly made available, through the Assistant Director, H. M. Geological Survey Office, Leeds, by the North Eastern Division of the National Coal Board of Great Britain. Accurately localized cores were used from the following borings: N.C.B. Durham off-shore No. 3; N.C.B. 5. Mainsforth "D"; N.C.B. Ferryhill No. 2; Garmondsway "D."

EXPERIMENTAL METHOD

The methods employed have been fully described elsewhere by the author (11, p. 461; 14, p. 883; 12) and need only brief mention here.

The rock is crushed and carefully ground and from it the fraction heavier than Bromoform (s.g. 2.9 gm cc) is extracted by centrifuging in that liquid. This fraction is predominantly composed of sulfide minerals. If necessary, residual light mineral material may be removed by standing for some days in warm hydrofluoric acid, but any organic material adhering to the spheres is not thus removed. The sulfide minerals may next be dissolved by the action of warm concentrated nitric acid and, from the resulting liquid, after careful washing free from acid, a suspension of micro-organisms released from the sulfide may be concentrated by final centrifuging. Permanent mounts in glycerine on microscope slides are then made. Alternatively, the process of solution of the sulfide from individual grains or small clusters may be observed through the microscope by using a special cell previously described (14, p. 884). All this preparation is carried out carefully under conditions allowing for the exclusion of all contamination, with freshly distilled water only being used throughout.

DESCRIPTION OF THE SULFIDE MINERAL FRACTION

Kupferschiefer.—Samples of the sulfide obtained from the disaggregated rocks were mounted on slides for microscopic examination by transmitted light, while a small number of polished sections of the original rocks were made for comparison. The latter method is that used by most earlier observers. In the grain mounts, as would be expected, simple spherical to oval, 4 to 15 μ spheres (Kieskügelchen) were abundant. The surface of most individual spheres specially examined at high magnification and under reflected light showed framboidal texture, a characteristic speckled appearance, described by Rust (1, p. 30–31). In polished sections (Figs. 1, 2) this could be seen to occur internally too, the sphere being composed throughout of large numbers of individual sulfide crystals, mostly idiomorphic. Instances were observed where in part a continuous mass of sulfide formed part of the sphere but whether the mass was of one homogenous mineral was not determined. Schouten has shown the possible range of secondary structures in massive spheres from the growth of sulfide between the primary sulfide to the complete replacement of the primary sulfide. Although occasional clusters were seen in the grain mounts, as in the polished sections, no clear evidence of actual union of adjacent spheres was seen, nor were any bodies clearly seen resembling the Kiesklümpchen with their larger individual well-spaced sulfide crystals, although some unusually coarse Kieskügelchen spheres were present. The polished sections also showed some spheres of a much looser structure than average, as if held in the process of breaking up. No such bodies were found in the loose grain mounts but probably the proportion of very small grains of sulfide in the latter was greater than in the original shale, due to the disintegration of many forms of sulfide during grinding. Small individual

grains of the size 2 to $\frac{1}{2} \mu$ and smaller are scattered throughout the original shale as "dust." Besides the spheres and "dust," larger angular grains of sulfide were abundant in the sulfide concentrate. This material undoubtedly represents fragments of more massive sulfide to be seen in small lenses and veins in the original rock but now randomly crushed down.

In the present study no detailed determinations were made of the mineralogical composition of the spheres and other sulfides with respect to particular metals. As would be expected, copper and other metals as well as iron were taken into solution by the nitric acid used to dissolve the sulfide.

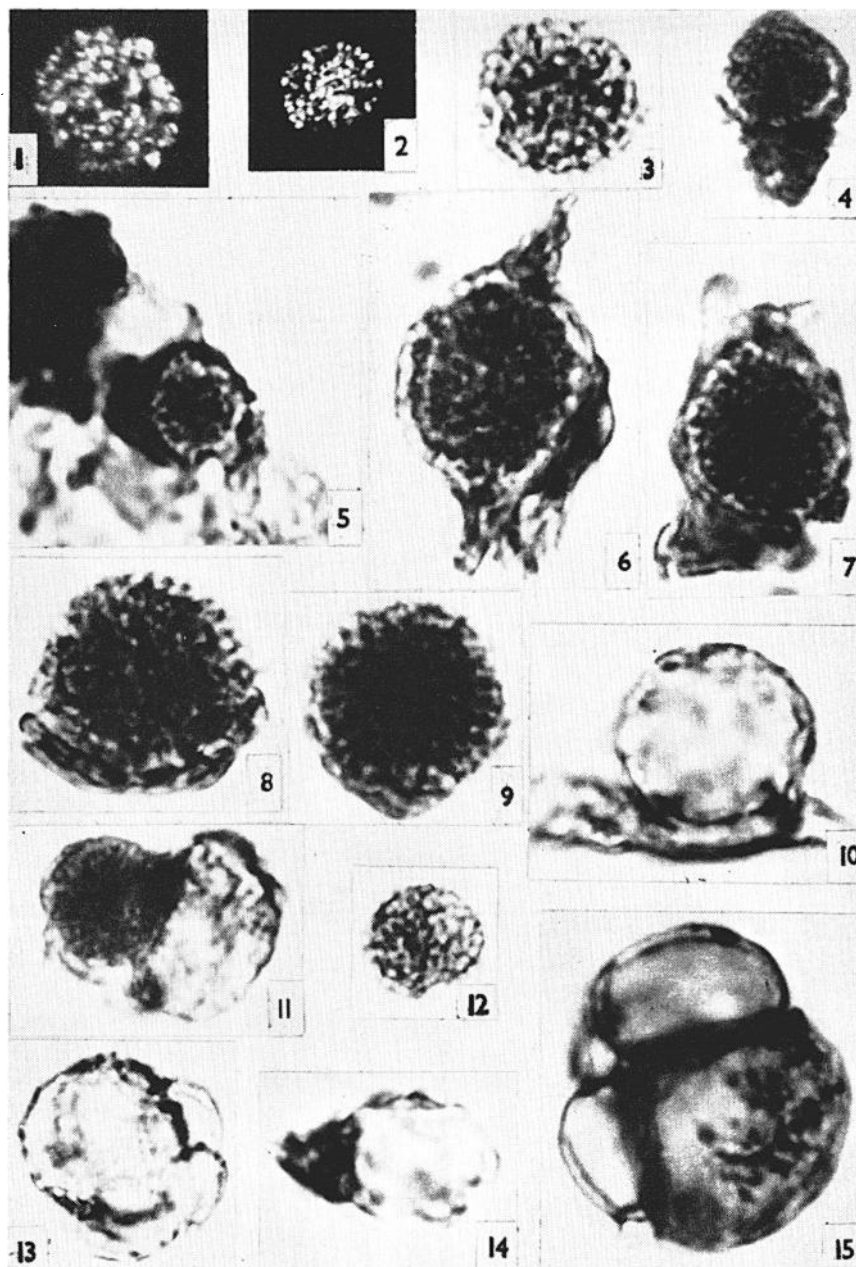
Although the use of hydrofluoric acid served well to remove any clay and other minerals that still remained adhering to the sulfide after centrifuging, it became clear in the course of the work that removal of all the bituminous material would seldom be possible although very intense grinding of the original rock reduced its proportion. Careful study showed that the sulfide spheres were firmly united with shreds and fragments and even quite large pieces of organic bituminous tissue from the shale, which were carried down by the sulfide into the heavy fraction. Tails of such material could also be seen, by transmitted light, on some spheres and, occasionally, very thin rims of translucent material although these were not visible by reflected light at the upper surface. Some of the spheres appeared to be associated with recognizable spore and pollen material and careful search revealed a small number of Hystrichospheres with grains of pyrite (not of Kieskügelchen type) borne internally. All these observations will be discussed later.

Marl Slate.—Qualitatively the sulfide forms in the Marl Slate closely resemble those of the Kupferschiefer, and spheres of Kieskügelchen type, 5 to 15 μ in diameter, and apparently all of pyrite, predominate and in some specimens are nearly as abundant as in the Kupferschiefer. Some grains of this type, however, also occur in a multiglobular form up to 20 μ or 30 μ across as if compounded from individual framboidal spheres. Evidence is given below that some of these are in fact unified bodies and not simply aggregated by chance on mounting the slide or by forming against each other in the original mud. Occasional Hystrichospheres with granular pyritic contents were also noted.

MICROFOSSILS RELEASED FROM THE SULFIDE

The bulk of the microfossils revealed by solution of the sulfide spheres and multispheres show the features of, and are assigned to, the species *Phyritosphaera barbaria* Love 1957 and the forms recognized as associated with that species (12) from Lias (Lower Jurassic) and Carboniferous strata. Only a little distinction can be made between the preparations obtained from the Marl Slate of north east England and from the Kupferschiefer of Germany.

The isolated spherical bodies, where free of any outer sac showed the characteristic form of being composed of a closely packed mass of small translucent grains $\frac{1}{2} \mu$ in size or less giving a microspinose to microgranular surface appearance (Figs. 3, 9, 12). Weak birefringence is shown by a few, the rest are isotropic. The birefringence is greatest immediately after clearing



FIGS. 1-2. Sulfide spheres (Kieskugeln) in Mansfeld Kupferschiefer. 1. Upper surface. 2. Section. Reflected light, oil immersion, approx. $\times 2,500$.

FIGS. 3-7. Microfossils released from sulfide spheres of Mansfeld Kupferschiefer. Transmitted light, oil immersion, approx. $\times 2,500$.

FIGS. 8-15. Microfossils released from sulfide spheres of Marl Slate of north east England. Transmitted light, oil immersion, approx. $\times 2,500$.

the sulfide with acid and decreases with longer contact with the acid and therefore it may be due to some initial product of the reaction, perhaps free sulfur.

The largest number of microfossils fall in the more complex form of *P. barbaria* (12) wherein the microspinose body is encased in a transparent to translucent outer sac, the latter commonly irregularly attached to or incorporated in other tissue material (Figs. 6, 7). Forms with some incomplete remnant of the sac are common (Figs. 4, 8). As was found in Lower Jurassic and Carboniferous rocks, instead of the typical spherical core body, in any of these instances, a smaller more irregular cluster of small granules may be found of material of similar appearance. Alternatively, rather loosely developed large forms of the microgranular to microspinose core body may be found. These were regarded as representing levels of development of the core material.

Where the spherical bodies are associated with large sheets of organic tissue (Fig. 5) they rest in what at first sight are closely fitting circular pits within its surface. In some cases, where the necessarily very delicate observation could be made, it has been confirmed that the lining of the pit continues over the upper surface of the body within, as a thin transparent skin. Many similar pits are seen devoid of core bodies, as might be expected from the inevitably rough treatment the material has received during preparation. When on the upper surface of the tissue under examination the pits appear to be completely closed on the lower side; but no empty pit with a closed upper surface as well has been positively confirmed. The organic tissue involved in this way is generally quite devoid of any regular structure or ornament to indicate its origin. Much of the abundant bituminous organic debris in these rocks is in a state of decomposition beyond that allowing recognition of its initial form. None the less some generally rounded masses, with *P. barbaria* on them, show suggestion of having been spore or pollen grains and some undoubted saccate pollen grains in the same state have been recognized.

A few pieces of tissue show a single open angular depression or group of depressions that are analogous to some idiomorphic form of sulfide crystal. Not appearing similar to the form originally described as *Pyritella polygonalis* (11, p. 434, & Pl. 33, Figs. 6, 7; 12) no organic significance is attached to them. It is probable that the crystal pressed into the soft organic tissue, perhaps during its formation, and during the laboratory preparation of the material the crystal carried the tissue down into the heavy fraction and subsequently was dissolved out.

The forms described above, and attributed to *P. barbaria*, are common to both the Kupferschiefer and the Marl Slate. In the latter, two further related forms of material occur but only a few examples have been seen from the Kupferschiefer. In the Marl Slate simple spherical sacs are (Figs. 10, 13, 14, 15) released from the sulfide, of a size and form comparable with those that encase the spherical core bodies. Some, observed very closely, seem quite unbroken and although most contain at least some small speck of material

inside, this is not a general rule. In the Marl Slate, too, compound bodies occur, clear to rough-walled cells, similar to the single ones but apparently in the relationship of parent to bud (Figs. 13, 15). Many of these too are found with core material present from small loosely granular clusters (Fig. 15) to well developed spherical microspinose or microgranular bodies (Fig. 11). The single and compound saccate bodies resemble those found in the same circumstances from pyrite spheres and multispheres of lower Jurassic and Carboniferous age, and are to be included under the saccate forms associated with *P. barbaria* (12).

Current study of recent sediments that contain abundant pyrite spheres suggests that the body *P. barbaria* represents the fossilized remains of a saprophytic organism that develops randomly within appropriate cavities in decomposing plant tissues. The pyrite then appears to form there rapidly. This means that sac formation is not an activity in the life of the micro-organism. This is discussed further by Love and Murray (13). The particular problem raised by the purely saccate forms remains to be solved.

The few pyritized Hystrichospheres principally assignable to the Michrystidia present aspects that distinguish them from the above group. The sulfide in them does not take the form of spheres and commonly is as little as one or two grains in an otherwise normal microfossil. On solution the pyrite grains reveal individual small organic cells; although when in a loose cluster these resemble the supposed early stages of *P. barbaria* they never take up the full development of the sphere characteristics of the latter. On these grounds, a distinction from *P. barbaria* is maintained; the material is more like that found within pyrite of animal tests of various ages (12). It will be the subject of communication elsewhere.

SULFIDE-ORGANIC MATTER RELATIONSHIP IN THE SPHERES

Some comment must now be made on the physical relationship of the sulfide to the range of organic structures recorded as associated with *Pyritosphaera barbaria*. Little can be added, however, to earlier discussions because with the equipment and experience available it has not been found possible to prepare polished sections clearly showing the relationship of sulfide to organic tissue, either around the whole 5 to 15 μ sphere or around the boundaries of the individual small grains whose light reflecting quality is very high. This lack, however, is relieved by the results of examining mounts of separated spheres and of watching the solution of individual spheres.

In the first place, there appears to be some justification in regarding the form of *P. barbaria* (the microspinose to microgranular core body) as being composed of many cells of organic tissue, each one of which is tightly filled with an individual crystal of sulfide. At least in the case of pyrite this is in the form of an octohedron or cube. During solution, the position of each crystal is left as the position of one small cell. The cells normally remain tightly united and they do not fall apart in this process. The outer members govern the external surface texture and very close examination can in some instances reveal the latter to be composed of the outward pointing angles of

crystal-shaped chambers, but this detail is normally seen in a generalized way over the whole sphere. The organic wall to each such $\frac{1}{2}$ to $2\ \mu$ cell is extremely thin and has not been observed in a polished section. The latter does however reveal what is fairly obvious from the solution study, that the regular grain structure, and so presumably the cell structure, generally continues right through the sphere. This is also to be observed in both Schneiderhöhn's and Schouten's photographs.

The presence of an outer sac, often proved during solution and sometimes detected on solid isolated sulfide spheres, does not obscure the framboidal appearance of the Kieskügelchen, and this is in keeping with its almost complete transparency after the sulfide has been removed, when it is frequently only detectable because it gives the apparently granular to microspinose core body a smooth equatorial outline. Possibly, too, it is initially thinner than after attack by nitric acid during which it might swell. In a simple sphere there may well be no pyrite external to this thin membrane for in many regular sulfide spheres no actual contraction accompanies the early stages of watched solution before the thin skin becomes apparent due to the solution of sulfide within it. The sac has been detected on a number of isolated spheres observed at high magnification in transmitted light, but would generally tend to be obscured by the bright refraction ring around the black sphere. In many cases trailing organic tissue can be seen, of the kind shown in Figures 4 and 6. In the cases where solution immediately produces a recognizable microspinose to microgranular *P. barbaria*, it is possible that in grinding the outer sac has been stripped from the sphere. On the other hand, (with the exception of the Hystrichospheres previously described) even when a sphere turns out to be a sac containing a small granular core body, only partially filling it, it does none the less generally seem to have been full of sulfide. This can be seen to dissolve away leaving a hollow cavity beside whatever core body there may be. This applies also to the multi-saccate bodies. This finding is borne out by polished surface studies, which do not reveal a class of grain internally hollow. If the specimens seen without any core material are truly undamaged and did not lose organic content during solution, then their relationship with sulfide presents an unsolved problem. It is not held that all spheres in these preparations were restricted to the size of their sac for it is not difficult to envisage further sulfide being precipitated externally and crystallizing on the outer side; certainly evidence of extra sulfide has been seen during solution.

It has been borne in mind, during the observations, that many spheres of sulfide as now seen are the products of one or more stages in a progression of replacements of one sulfide mineral by another. The question naturally arises, given the organism-sulfide relationships suggested above, of what effect there is likely to be on this when one sulfide is replaced by another under the conditions accompanying such metasomatism. Often evidence of the nature of a system can be deduced from the way the system reacts to external influences. Little, though, is derived from study of the reports by Schouten and others that during replacement some differentiation may occur between core and rim and that new sulfide may enter interstitially between the grains in a sphere. Two factors prevented this and other replacement data

from being of use, for it was neither possible to know the precise mineralogical composition of a sphere whose solution was about to be watched nor to segregate batches of spheres of a uniform composition. The organic tissue of those isolated spheres whose solution has been watched has not shown disruption as if due to earlier replacement of sulfide but indeed none might be expected for replacement would occur particle by particle through what is clearly a permeable structure. As noted earlier, however, in polished sections of rock, some loose clusters of sulfide grains could be seen that might have been interpreted as being *Kieskügelchen* spheres falling apart. Such grains have been figured by Schouten (22, Fig. 80). In this case, the masses would not remain as complete spheres on disaggregation of the rock to give a sulfide concentrate. There is no knowing, indeed, the number of *Kieskügelchen* that simply remained as scattered unit crystals after this process, and further how many did so as a result of changes during geological processes. Certainly some such minute grains did yield a tiny cell of organic tissue on solution while many others did not.

FORMATION OF THE PRIMARY SULFIDE

Syngenetic origin of the sulfide spheres was prominently expounded by Schneiderhöhn (20, 21) who regarded them as having formed from blobs of the mixed gels of iron, copper and other sulfides all produced contemporaneously on the sea floor. The spheres represented sulfur bacteria mineralized with sulfide ores, and the points or crystals in the spheres were individually mineralized sulfur granules in the bacteria. Schouten, however, demonstrated the evidence for the primary sulfide composition of the *Kieskügelchen* having been that of pyrite with subsequent replacement, in which "chalcocite, bornite, chalcopyrite and, less commonly, galena and sphalerite are found as pseudomorphs after the original pyrite crystals" (22, p. 377). In general the replacement was regarded as having been by the locally predominant mineral at the relevant stage of metasomatism. Part of the evidence for the replacements lies in the common preservation of the individual grain formation characteristic of the pyrite spheres, or alternatively, the structure may appear to have been lost in the core of the sphere, with the outer parts still showing the original structure. Siegl (25) regarded pyrite as the primary sulfide in the spheres (although itself replacive, see below), while Neuhaus (15), having the same opinion for both *Kieskügelchen* and *Kiesklümpchen*, noted that other sulfides might fill the interstices of the pyrite grains before replacing the pyrite.

Schouten (23, p. 529), admitted "that the arguments for a syngenetic deposition of FeS_2 , with the aid of sulphate reducing bacteria, dominate" but remained unconvinced of the existence of fossil sulfur bacteria. Also unconvinced that the iron sulfide, as the spheres, was syngenetic he elsewhere suggests (22, p. 377) that the pyrite spheres might have arisen from replacement of carbonate by iron sulfide. "In the copper shale a most remarkable conformity exists between the great variety of shapes, dimensions and mode of distribution of carbonate and sulphide bodies. From this it follows that

pseudomorphism of carbonate by sulphide may be presumed." While this refers mainly to lenses and irregularly shaped grains, an instance is cited (op. cit. p. 379) from the Kupferschiefer shale of "accumulations of smaller carbonate grains approaching more or less globular forms" with sulfide grains of similar form adjacent. In the case cited, however, the sulfide is bornite and galena and the framboidal form does not appear to be present. A similar point is raised by Siegl (25, p. 350-51) who finds the carbonate groundmass present in some of the rocks in question here to be formed of small grains comparable in size with the Kieskugelchen, and regards the latter as having been formed by pyritization of the former. No detailed description of the calcite grains was given. Siegl's discussion is referred to by Deans (3, p. 350).

Relevant to this discussion is an example considered by Rayner (6, p. 202-3) in a description of a shelly limestone from the Coal Measures of England. In this she finds minute spheres of carbonate, probably ankerite, and limonite, and also pyrite; and the association is compared with that noted above by Siegl. In the case of the English shelly limestone the present author was, by the kindness of Dr. Rayner, permitted to examine a sample of the rock and its thin sections. Without doubt the pyrite spheres were framboidal and on separation and chemical treatment they yielded the range of organisms described by Love (12), and identical with those described in the present paper. On the other hand, the granular carbonate and limonite, free of pyrite, were demonstrated to contain no such organic matter. They were completely soluble in hydrochloric acid except for a small amount of mineral material, probably clays. In this case then, there can be no possibility of conversion of the carbonate into pyrite spheres as it would be necessary to introduce the forms of *P. barbaria* as well as the pyrite. Furthermore, at very high magnification there was little similarity between the small 4 to 14 μ sub-angular grains of calcite, each a single-crystal grain, and the framboidal Kieskugelchen of the same general size but with their own very characteristic regular form. In the work on the Kupferschiefer described in the present paper, it was not found that calcite grains of this size are common in thin sections, and furthermore it was not found possible to obtain the micro-organisms described by use of cold or warm dilute hydrochloric acid on the rock. This means that none of those bodies were included either in calcite or other carbonate grains. Indeed, following the instruction of Schouten (22, p. 376) "As an axiomatic principle it may be said that the internal structure of the host governs the final result" it is difficult to understand how the framboidal texture of the pyrite Kugelchen, so tenaciously retained through all but the last stages of replacement, should itself have arisen from diagenetic replacement of rather vague and simple calcite grains.

We return then to the dominating conclusion reached by Schouten of pyrite having been the primary mineral in the spheres, with the added knowledge of the particular micro-organisms occurring within them to emphasize their syngenetic origin.

In the sense used by earlier authors on the Kupferschiefer problem, the pyrite spheres would have been termed syngenetic, in contrast to an epigenetic origin from hydrothermal solutions after lithification of the sediment.

More strictly, however, within the sequence of formation of the sediment it is clear that much of the pyrite was not present until after finely comminuted plant debris and the clay matrix had been deposited as a mud rich in iron. As part of the earliest process of diagenesis, authigenic growth of sulfide mineral then occurred from iron and, presumably, sulfate-sulfur from that part of the water enclosed within the mud. Until a point was reached such that the system became closed in any part of the mud then the formation of iron sulfide there would continue as long as sulfate was available from the overlying water and iron from the same source or from adsorption complexes with organic matter already in the sediment. Indeed the distinction between syngenetic and early diagenetic is not easy to draw, and it is perhaps allowable to continue using the term syngenetic here for the primary sulfide in simple distinction from much later diagenetic or epigenetic formation of the secondary sulfide.

FORMATION OF THE SECONDARY SULFIDE

At some time subsequent to the formation of the iron sulfide in the spheres and elsewhere scattered throughout the sediment, there was an influx of copper, lead, zinc and other metals to replace the iron. In the literature, the formation of the ores of these metals, whether syngenetic or epigenetic is inextricably bound up with the origin of the iron sulfide. Indeed it is not possible to discuss the relationship between these primary and secondary sulfide minerals without bringing in wider issues of ore genesis as a whole in the Kupferschiefer and elsewhere.

In general some cases of mineralization appear to accompany dark shale sediments and their particular sedimentary environment in a manner that makes their origin in them, if due to epigenetic hydrothermal activity, difficult to understand. At the same time, and the Kupferschiefer is typical of this with regard to the provenance of the metals, it is equally difficult to reconcile the great amounts of metal involved with any but an original magmatic source. Again the regular concentration of the ores in terms of area is a factor tending to invoke this form of origin. Although as Deans (3, p. 347) points out, the Kupferschiefer type of mineralization can be recognized in north east England, it is in central Germany that copper, zinc, and lead are most abundant.

In their paleogeographic setting, the Kupferschiefer and the Marl Slate were the first fine deposits in the Lower Zechstein sea after the strong transgression that terminated the continental Rotliegende period. This submerged an area from the north east of England across Germany to Russia, and as shown by Gregory (8, p. 22-3) the Kupferschiefer itself formed in an area receiving drainage from the Harz, the Erzgebirge and the Thuringerwald, all uplands well known for their earlier mineralization. Gregory concluded that a series of lagoons might be envisaged marginal to the main Zechstein sea and perhaps partially enclosed from it. The most thorough study of the conditions of deposition of the Kupferschiefer there is due to Pompeckj (17), summarized in English by Schuchert (24). The first deposit in these lagoons of the whole Zechstein succession was the Kupferschiefer. The latter author in

comment, upholds Pompeckj's conclusions that the Kupferschiefer was formed in a foul-bottomed sea area with upper oxygenated waters abounding in plankton and in drifting plant material, the latter often yielding excellent fossils and being indicative of nearness to land from which only the finest terrigenous sediment was being washed. All this, on sinking to the bottom, gave the source of the rich bituminous content (6–20%) of the fine grained dark mud in process of becoming the Kupferschiefer. Although much of the fine-grained bitumen is now unrecognizable, it is not entirely structureless for, in suitable micro-preparations, abundant plankton and plant material can be recognized. The Kupferschiefer differed from later Zechstein deposits also in that brachiopod and molluscan life was very sparse and few scavengers were present. Only one species of fish is common, *Palaeoniscus frieslebeni*; most of the others were regarded by Pompeckj as having been washed in from rivers rather than having been indigenous to the area of deposition. In general the majority of life was regarded as having occurred in the well aerated upper waters and although some bottom feeding took place, abundant bottom life was not possible because of the lack of oxygen. The depth of the Kupferschiefer sea was regarded by Schuchert as not exceeding 600 feet, much less than that of the Black Sea, but not less than 300 feet, the lower limit of aeration by storm waves. The Kupferschiefer deposits show no periodic change to aerated conditions, nor any disturbance of their fine lamination.

Explanation of many of the extensive and commonly thick series of dark shale in the geological column in terms of conditions known at the present has always been difficult. Persistency of stagnation, however, is partly explained by the concept of the barred basin (29) with circulation of water greatly restricted. Indeed if the bar does not reach to the surface then denser saline water entering from the open sea and forming a heavier underlayer may set up an extra resistance to thermal convection. Strøm's studies (27) on Norwegian fjords show the feasibility of reducing conditions having been maintained unbroken at least for many centuries. However, nowhere at the present time, except in the particularly deep Black Sea, do we find the great areas of persistent stagnancy that must have occurred in the early Zechstein sea, or in the Lias sea later covering much the same area of Europe. The recognition of the continuation of the Kupferschiefer sea into Northern England indeed suggests that a very large marginal sea area rather than small lagoons must have been the case. Following Strøm, however, it may be suggested that present day oceans are geologically abnormal in their extensive underflows of cold aerated waters from polar sources. Both a different polar climate and the Mediterranean situation of the Tethyan sea, of which the Kupferschiefer and Lias dark shale areas were marginal, may have been factors in favoring the persistent stagnancy evidenced by these deposits.

In this environment it is now necessary to consider the role of the micro-organisms described in this paper in relation to earlier ideas on the formation of the sulfide of the spheres, and on the provenance of the metals themselves. There are the contrasting views, already mentioned, of Schneiderhöhn in favor of simultaneous precipitation of iron, copper, lead and zinc sulfides, and of Schouten emphasizing the sequences of metasomatic replacement of iron

sulfide in the spheres by epigenetic solutions of copper, lead and zinc. The timing of the latter, however, has been questioned by Deans (3, p. 350) who suggested on the balance of evidence that the replacement was diagenetic rather than epigenetic. The fact that the primary sulfide was pyrite, even if later pseudomorphed by the others, is not detrimental to the diagenetic interpretation for it has been shown by Love and Murray (13) how early this mineral can appear in foetid sediment.

Two theories have been employed by earlier writers to account for the source of all these metals for syngenetic deposition in the Kupferschiefer sea.

1. Leaching from pre-existing humus soils or crusts of dried-up playa deposits. These may have been of long standing, themselves enriched by the action of percolating waters on metalliferous rocks beneath.
2. Direct weathering from deposits associated with ores or eruptive rocks, perhaps having occurred over a long period of time with the accumulated solutions lying as deep groundwaters before rising again, in a period of greater rainfall to enrich the Kupferschiefer sea.

In the case of transport by land drainage from such sources into the sea, a mixed load of metals must be considered. Richter (18) from the resulting regular distribution of metals in the water, suggested that varying water conditions would control deposition of the metals copper, lead and zinc. They would be precipitated in this order in the most sulfurous waters but in fresher streams copper might remain in solution while zinc precipitated as carbonate or hydroxide. If the chemical environment in the immediate locality changed, then this metal might become available for diagenetic replacement. Richter did not discuss the formation of iron. Deans, however, (3, p. 350) considering the vertical sequence of strata noted that the abundance of copper, lead and zinc is what, in terms of potential series and general order of solubility, might result from processes of sedimentary deposition. He suggested it was an early phase of the later established succession of salt precipitation in the Zechstein controlled by concentration. Yet with the metals present together after entry from a common source it is difficult on theoretical grounds to explain the selective deposition of iron as sulfide leaving the residual metals to be concentrated. Despite the conditions of restricted circulation present, it is not considered probable that, following Goldberg (7, p. 258-9), falling microplankton and other fossil remains would play a significant part in concentrating the more common metals, particularly iron, in the sediment.

Looking more closely at the normal state of these metals it may be noted that Hutchinson (10, p. 708 et seq. & p. 811) emphasizes the very small concentrations of free ions of iron, copper and other metals normally found in the water and sediments of rivers and lakes, for most of these metals are present in non-ionic association with colloidal material both organic and inorganic. The latter is commonly clay and ferric hydroxide. In this state then they are carried into the sea and the more readily precipitated on encountering the saline water. Special conditions hold, however, in the hypolimnion of stagnant lakes and, presumably, marine waters. Here, where the oxygen content

has fallen to very low levels (10, p. 716–720 and Fig. 215) a considerable amount of ferrous iron may be found in solution. It appears to be drawn into an ionic state, simple or complex, from adsorbed, non-ionic iron in the sediment; and this process would continue, to balance an equilibrium, as iron sulfide is precipitated. Hutchinson (10, p. 762) demonstrates the virtual undetectability of copper as simple ions in the presence of ferrous sulfide once the latter is present. Considering the effect of excessive concentration of iron on the equilibria (and, although seldom mentioned, iron is predominant among the metals of the Kupferschiefer) he shows (10, p. 762) that in the presence of ferrous sulfide “no limnologically possible concentration of ferrous iron could, by decreasing the concentration of the sulfide ion, ever permit the presence of analytically detectable amounts of copper,” and only a slightly greater extent of lead and zinc, in solution. Nevertheless Hutchinson notes from equilibrium data that beneath this limit, and probably well below a toxic concentration of copper ion in the water, transfers of copper from the adsorbed non-ionic condition to replace iron in ferrous sulfide would proceed steadily. Some mobility perhaps occurs also in complex form with an amino compound (op. cit., p. 815).

With a different starting point of a directly submerged Rasenkupferstein oxide crust, postulated by some authors, again reducing conditions in the bottom waters might allow preferential leaching of iron, to be redeposited as sulfide before its replacement more slowly by other metals entering into this more stable phase. Consideration for the first time for the Kupferschiefer, however, of trace elements which are regarded as characteristic of ancient soils and sapropels led Deans (3, p. 350) to allow the possibility of ancient soils as a source of metals. But in the light of more recently elaborated theories it may be possible to go further than Deans in suggesting more than a single source for the major minerals present, to give a more satisfactory explanation of the differentiation of the metals in the formation of the sulfides.

Iron is normally an important constituent of water rich in humic constituents and, mostly precipitated as sulfide as a result of biological sulfide reaction, iron is also normally abundant in the dark foetid muds of stagnant basins rich in organic matter. But with regard to other heavy minerals in the Kupferschiefer Dunham (5, p. 288) points out that the suite is “not normally formed in quantity even in a highly evaporative lagoon.” For this reason in the case of the Zechstein sea he envisages the iron as having been brought into the stagnant waters by rivers rich in humic and plant material, and iron sulfide as forming under microbiological influence in the bottom mud; then “springs connected with the hydrothermal mineralisation then in progress in the Harz, the Thuringerwald, the English Pennines and possibly elsewhere broke through into the Lagoon discharging a large volume of mineralised waters left over after producing extensive epigenetic deposits in those districts.” “The metals were deposited either bacterially or by reaction with iron sulphide already present.” From consideration of the replacement series already discussed, of these two possibilities, replacement of pre-existing iron sulfide was the principal reaction that occurred. With regard to the time of this reaction Dunham refers to Westoll’s evidence (28; 3, p. 351) that copper

and other metals entered into the head structures of certain fossil fishes in the Marl Slate of Northern England early in the decay process before their delicate structure could be crushed beneath the accumulating sediment. Davidson, however (2), rejecting modern "syngenetic" hypotheses on the origin of strata-bound polymetallic deposits, suggests that the fish structures might earlier have been represented by pyrite of which the iron was later replaced by other metals. There appears to be no published evidence on this point. Dunham even suggests that a mass killing of fishes may have occurred in connection with the emission of the mineralized waters. None the less it is probable that the metals would have been very rapidly absorbed non-ionically within the sediment, and the possibility of prolonged toxicity has already been dismissed. It remains to be stated whether the emission was relatively sudden or a prolonged seepage. There seem to be no horizons of the Kupferschiefer richest in all metals as if representing the maximum outburst, and explanation is required of the trends of both vertical and horizontal distribution of the ore metals such as noted by Deans (3, Figs. 85, 86) even if some subsequent transport has occurred along Rücken and into over and underlying strata. Oftedahl (16, p. 17) has suggested that the submarine exhalations may have been gases, associated with magmatic activity. This would be difficult to demonstrate, but the associations recognized in some instances by Stanton (26, p. 24) between submarine extrusive rocks and the accumulation of sulfide ore is certainly not applicable to the case of the Kupferschiefer, for no such volcanic rocks are known for this period although lavas and tuffs occur in the underlying Rotliegende.

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REFERENCES

1. Bastin, E. S., 1950, Interpretation of ore textures: *Geol. Soc. America Mem.*, v. 45.
2. Davidson, C. F., 1962, On the origin of some strata-bound sulfide ore deposits. Discussion: *ECON. GEOL.*, v. 57, p. 265-273.
3. Deans, T., 1948, The Kupferschiefer and the associated lead-zinc mineralisation in the Permian of Silesia, Germany and England: *Rept. 18th Intl. Geol. Cong., Gt. Britain*, 1948. Pt. vii, p. 340-352.
4. Dunham, K. C., 1960, Syngenetic and Diagenetic Mineralisation in Yorkshire: *Yorks. Geol. Soc. Proc.*, v. 32, p. 229-284.
5. —, 1961, Black shale, oil and sulfide ore: *Adv. Sci.*, v. 18, No. 73, p. 284-299.
6. Eagar, R. M. C., and Rayner, D., 1952, A non-marine shelly limestone and other faunal horizons from the Coal Measures near Wakefield: *Leeds Geol. Assoc. Trans.*, v. 6, p. 188-209.
7. Goldberg, E. D., 1954, Marine geochemistry. 1. Chemical scavengers of the sea: *Jour. Geology*, v. 62, p. 249-265.
8. Gregory, J. W., 1930, The Copper-Shale (Kupferschiefer) of Mansfeld: *Inst. Min. Met. Trans.*, v. 40, p. 1-55.
9. Grondijs, H. F., and Schouten, C., 1937, A study of Mount Isa ores: *ECON. GEOL.*, v. 32, p. 407-450.
10. Hutchinson, G. E., 1957, *A Treatise on Limnology*: v. 1, John Wiley & Sons, New York.
11. Love, L. G., 1957, Micro-organisms and the presence of syngenetic pyrite: *Geol. Soc. London Quart. Jour.*, v. 113, p. 429-440.
12. —, 1962, Further studies on micro-organisms and the presence of syngenetic pyrite: *Palaeontology*, v. 5.

13. Love, L. G., and Murray, J. W., 1962, Biogenetic pyrite in Recent sediments of Christchurch Harbour, England: *Am. Jour. Sci.*
14. Love, L. G., and Zimmerman, D. O., 1961, Bedded pyrite and micro-organisms from the Mount Isa Shale: *ECON. GEOL.*, v. 56, p. 873-896.
15. Neuhaus, A., 1940, Über die Erzführung des Kupfermergels der Haaseler und der Gröditzter Mulde in Schleisien: *Zeitschr. f. angew. Min.*, v. 2, p. 304-343.
16. Oftedahl, C., 1958, A theory of exhalative sedimentary ores: *Geol. Foren. Forh.*, v. 80, p. 1-19.
17. Pompeckj, F. J., 1914, Das Meer des Kupferschiefer: *Branca-Festschrift*, Leipzig, p. 444-494.
18. Richter, G., 1941, Geologische Gesetzmässigkeiten in der Metallführung des Kupferschiefers: *Arch. Lagerst.-Forsch.*, v. 73.
19. Rust, G. W., 1935, Colloidal primary copper ores at Cornwall Mines, Southeastern Missouri: *Jour. Geology*, v. 43, p. 398-426.
20. Schneiderhöhn, H., 1923, Chalkographische Untersuchung des Mansfelder Kupferschiefers: *N. Jahrb. f. Min.*, B47, p. 1-38.
21. —, 1926, Erzführung und Gefüge des Mansfelder Kupferschiefers: *Metall und Erz*, v. 23, p. 143-146.
22. Schouten, C., 1946, Some notes on micropseudomorphism: *ECON. GEOL.*, v. 41, p. 348-382.
23. —, 1946, The role of sulfur bacteria in the formation of the so-called sedimentary copper ores and pyrite ore bodies: *ECON. GEOL.*, v. 41, p. 517-538.
24. Schuchert, C., 1915, The conditions of black shale deposition as illustrated by the Kupferschiefer and Lias of Germany: *Am. Phil. Soc. Trans.*, v. 54, p. 259-269.
25. Siegl, W., 1941, Zur Genesis des Kupferschiefers: *Min. Petr. Mitt.*, v. 52, p. 347-362.
26. Stanton, R. L., 1960, General features of the conformable "Pyritic" Orebodies. Part I—Field Association. *Can. Inst. Min. Trans.*, v. 63, p. 22-27.
27. Strøm, K. M., 1939, Landlocked waters and the deposition of Black Muds: *Recent Marine Sediments* Ed., P. D. Trask, London, p. 356-372.
28. Westoll, T. S., 1943, Mineralisation of the Permian rocks of S. Durham: *Geol. Mag.*, v. 80, p. 119-120.
29. Woolnough, W. G., 1937, Sedimentation in Barred Basins and Source Rocks of Oil: *Am. Assoc. Petr. Geol. Bull.*, v. 21, p. 1101-1157.